BREJESH BALAKRISHNAN

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EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering | CGPA: 8.54/10 **Jun 2020 - May 2024**

College of Engineering Guindy, Anna University, Chennai, Tamil Nadu, India

Relevant Coursework: Artificial Intelligence and Machine Learning, Probability and Random Processes, Linear Algebra, Signals and Systems, Data Structures, Database Management Systems, Digital Systems, Microprocessors, Embedded Systems, Control Systems, VLSI Design

ACADEMIC PROJECTS & RESEARCH WORK

ML Based Identification and Navigation System for Visually Impaired Users **Dec 2023 - May 2024**

Final year project | Guide: Dr. T. Sridarshini, Department of ECE | Team of 4

- Developed a wearable navigation system for the visually impaired on a Raspberry Pi 4, combining an HC-SR04 ultrasonic proximity sensor with camera-based object detection; converted detections into spoken alerts with a 2 second debounce so the user is not flooded with speech.
- Obtained a 3.3% accuracy increase over baseline SSD on COCO by adding a Convolutional Block Attention Module and replacing IoU with DIoU for box regression, reaching 0.96 accuracy, 0.963 precision, and 450 ms inference, with the largest gain on small objects.
- Benchmarked the design with six alternatives from a four-paper literature survey (YOLOv5s variants, Viola-Jones, neural baselines) and against ultrasonic-only sensing, which covers 2 cm to 2 m at 40 degrees versus 30 cm to 6 m at 78 to 108 degrees for the camera pipeline.

Tools/Techniques Used: Python, OpenCV, PyTorch, SSD, CBAM, DIoU, COCO, Raspberry Pi 4, gTTS

Agent GovBench: Benchmarking Guardrails for Tool-Using AI Agents **Aug 2026 - Oct 2026**

- Built an open benchmark to measure the capability of guardrails that stop a tool-using agent from being misused on Google Cloud (Agent Development Kit, 9 tools across 5 stub MCP servers), with 999 synthetic scenarios, 45% of them proper requests that only appear risky, and attack success scored from the tool-call log by a deterministic check rather than an LLM judge alone.
- Compared guards on similar test calls and found Model Armor at its MEDIUM injection threshold blocked 29.8% of malicious calls while flagging 27.1% of benign ones, against 94.7% and 0.4% for a Gemini Flash guard; with the agent in the loop.
- Distilled Gemini 3.1 Pro labels into a LoRA-tuned Gemma 4 E4B guard on leakage-checked splits that blocked 62% of attacks at 8% false positives on 1,566 real agent calls (98% at 3% for Gemini), with tuning reducing false positives at unchanged recall by almost half. Published the harness, sanitized data and model with bootstrap intervals and stated limits, and logged Agent Gateway governance as not tested rather than estimated.

Tools/Techniques Used: Python, Google Agent Development Kit, Gemini, Gemma 4, LoRA, Model Armor, MCP, Cloud Run, Terraform, Hugging Face Hub

Agentic RAG Workbench: Retrieval Backend with Evaluation Harness **Sep 2025 – Jan 2026**

- Built an 8 node LangGraph pipeline with conditional routing (query normalization, multi-turn contextualization, dense Pinecone retrieval, bounded corrective retrieval, web fallback, grounded generation, faithfulness verification), served over FastAPI with SSE streaming.
- Tested a hand-labelled golden set on recall@k, MRR and nDCG@k in a A/B scenario to reject a hosted reranker, it came out net negative (nDCG@3 down 0.057, latency up 435 ms), so it was disabled on the evidence rather than shipped on assumption.
- Designed a two-threshold retrieval gate with a programmatic decline path so the model is never asked to answer from empty context, covered by 343 tests and a hash-pinned CI lockfile.

Tools/Techniques Used: Python, LangGraph, Pinecone, Groq, FastAPI, Docker, Prometheus, Streamlit

Customer Churn Prediction: End to End MLOps with SHAP Grounded Explanations **Nov 2025 - Mar 2026**

- Built a leakage-safe scikit-learn pipeline on the IBM Telco dataset (7,043 records, 26.5% churn), benchmarked it against six models under 5-fold stratified cross validation on PR-AUC, tuned XGBoost with Optuna, and from an 5:1 FN to FP cost set the decision threshold at 0.174.
- Obtained test score of PR-AUC 0.66, ROC-AUC 0.85 and recall 0.87 with an optimism gap of 0.01. Additionally, a two-tier faithfulness check on the SHAP-grounded explanation layer exposed a real grounding bug and lifted faithfulness from 0.72 to 0.90.
- Enforced a single-source Pandera data contract across training and serving so the test suite fails if the API schema drifts, with Evidently drift monitoring and an MLflow champion/challenger registry.

Tools/Techniques Used: Python, XGBoost, Optuna, SHAP, MLflow, DVC, Evidently, Pandera, FastAPI, Docker, GitHub Actions

SafetyOps Copilot: Event-Driven Workplace Safety Triage Platform **Mar 2026 – July 2026**

- Fine-tuned DistilBERT for text-based incident-severity classification on 273k real regulatory outcome labelled MSHA mine-safety narratives, compared against a rule-based baseline to get a score of macro-F1 0.78 against 0.34 and high-severity recall 0.88 against 0.16.
- Built the wrapping platform in a FastAPI service on a Redis Streams bus with consumer groups and a shared PostgreSQL store, with idempotent processing, exponential-backoff retries and a dead-letter queue with replay.

Tools/Techniques Used: PyTorch, DistilBERT, YOLOv8, FastAPI, Redis Streams, PostgreSQL, LangGraph, Evidently, Grafana

WORK EXPERIENCE

AI Engineer, Cloud202 Limited, United Kingdom (Remote) **Sep 2026 - Present**

- Audited governance and safety in Qubitz, *the company's platform for building AI agents on AWS*, against the 4 projects in AWS's sample AI agent factory, covering 15+ governance features.
- Designed a phased plan, as approved, to build governance controls into the platform and the agents it creates on Amazon Bedrock and Bedrock AgentCore.
- Researched Amazon SageMaker to identify features that can extend the company's enterprise offering across client projects, alongside 7 AWS trainings and accreditations in sales, technical, cloud economics and agentic AI.

Advanced Application Engineering Analyst, Accenture, Bengaluru, India **Jul 2024 - Mar 2026**

- Optimized a Java and Spring Boot financial services platform hosted on AWS REST APIs, cutting back responses slower than 5 seconds from roughly 80% to under 20% through profiling, SQL query tuning, and caching.
- Delivered 200+ story points across Agile sprints, Streamlining loosely provided requirements into shipped features alongside UI and QA counterparts, and presented sprint demos to client stakeholders.
- Maintained automated test coverage above 90% across unit, contract, and BDD suites focusing on establishing proper regression suites, and closed critical defects through timely root-cause analysis.
- Onboarded 3 new joiners onto the codebase and assisted with local setup, briefed them on the client and team's work, walked them through their first sprint and reviewed their code.

Intern, UnisLink, Chennai, India **Jul 2023 - Aug 2023**

- Researched natural language to SQL approaches and reviewed commercially available products, documenting the accuracy against complexity trade off along with the cost associated with it to shape the early product direction.
- Validated SQL Server replication and synchronization checks across large tables, then documented findings and handoffs for the team.

Analytics Intern, Yaltech Global Consulting, Tiruchirappalli, India **Jul 2022 - Sep 2022**

- Trained and cross-validated regression and classification models in Python and scikit-learn on structured business datasets, to build a car recommendation system based on it.
- Performed exploratory analysis, cleaning and feature preparation in Pandas, and presented the work and findings to non-technical stakeholders through Seaborn and Matplotlib visuals.

Intern, TREC-STEP Technology Business Incubator, Tiruchirappalli, India **Apr 2021 - Jun 2021**

- Conducted desk research and drafted sections in an EU Horizon 2020 proposal on Integrated Local Energy Systems and in a CIFAR Responsible AI proposal under the Canada Solution Network (STAR RAISE).
- Coordinated incubation activities including startup interactions and site visits for programs on innovation, climate, and gender inclusion.

ROLE OF RESPONSIBILITY & ACHIEVEMENTS

Junior Executive, Projects and Events, Robotics Club, College of Engineering Guindy **2022 - 2024**

- Conducted 2 hands-on workshops for junior students, one for building a line tracing robot and another combining the same electronics with machine learning.
- Mentored a 3-student team for a medical image classification and detection project, advising on methodology and tooling, clearing doubts, and guiding them for a review presentation with club advisers.

Executive Head, Events and Planning, AUSEC, College of Engineering Guindy **2022 - 2023**

- Led a 3-person team to run a full day digital marketing workshop for 150 to 200 participants, sourcing the speaker, coordinating with the guest team, preparing the material and issuing certificates, and ran a day long games stall that drew around 200 visitors.

Projects Team Member, CEG Tech Forum, Kurukshetra '23 **2022 - 2023**

- Built the two classification models behind AU Hitch Handler, an application for registering complaints about logistic problems around the university: one filtering toxic submissions, the second directing each complaint to the relevant department, benchmarking classical algorithms against a fine-tuned BERT model on both tasks.

Coordinator and Senior Volunteer, National Service Scheme Unit 11, CEG Campus **2022 - 2023**

- Organized Life Pot project as the sole organizer, sourcing earthen water pots from Kerala and placing over 200 of them across the campus and staff quarters so stray animals and birds had drinking water through the Chennai summer.
- Led the volunteer team on a hair donation drive that collected more than 50 donations towards wigs for cancer patients, organized first aid camps and alumni meets, joined a Besant Nagar beach cleaning with the Environmentalist Foundation of India, and mentored junior volunteers through camps and fresher orientation.
- Completed a seven-day residential camp at Thirupalaivanam, Tamil Nadu, painting village middle schools, cleaning and repainting the communal pond, planting and distributing saplings and organizing rallies for water management.

ACHIEVEMENTS

- Received a Letter of Appreciation signed by Dr. K. Rathnakannan, Staff Coordinator of CEG Tech Forum, for the machine learning contribution to AU Hitch Handler at **Kurukshetra '23**. **Apr 2023**
- Served as School Representative while maintaining top academic standing, ranking 1st (2019) and 3rd (2020) in school.

SKILLS

Languages: Tamil (Native), Malayalam (Native), English (Fluent), Hindi (Intermediate)

ICT: Python, SQL, Java, C/C++, MATLAB, PyTorch, scikit-learn, XGBoost, Transformers, LangGraph, LangChain, OpenCV, MLflow, DVC, Docker, GitHub Actions, Evidently, Pandera, FastAPI, Streamlit, Spring Boot, Angular, Pinecone, FAISS, PostgreSQL, MongoDB, AWS (SageMaker, Bedrock, Bedrock AgentCore Lambda, S3), Google Cloud (Agent Development Kit, Model Armor, Cloud Run), Oracle Cloud Infrastructure, Git, Linux, Power BI

Certifications: AWS Machine Learning Engineer Nanodegree (Udacity, 2025); Oracle Cloud Infrastructure Data Science Professional (Oracle, 2025); Advanced Data Science and AI Program (Logicmojo Academy, 2026); McKinsey Forward Program (McKinsey & Company, 2026); AWS Partner: Accreditation (Technical), AWS Partner: Sales Accreditation (Business) and AWS Partner: Cloud Economics Accreditation (AWS, 2026);Version Control with Git (Atlassian, 2023); CCNA v7 Introduction to Networks (Cisco Networking Academy, 2023)